

Aravind B

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Research Statement

Dedicated Machine Learning Engineer and Healthcare AI Researcher with over seven years of experience in applied ML, computer vision, and medical imaging and MLOps. Published author with a proven track record of architecting explainable deep neural networks, deploying containerized healthcare pipelines, and leading production-grade AI engineering initiatives. Seeking a PhD position to advance research in explainable medical AI and clinical decision-support systems while contributing to cutting-edge interdisciplinary research.

Education

- **MSc in Embedded Systems** | Distinction (70%) | Coventry University, United Kingdom | 2019
Thesis: Healthcare System Platform using Convolutional Neural Networks (DCNNs)
- **B. Tech in Electronics & Communication Engineering** | Distinction (70%) | Audisankara College of Engineering, JNTU Anantapur | 2014
Capstone Project: Classification of Microcalcification using Shearlet Transform and SVM
- **Intermediate** | 94.2% | Narayana Junior College | 2010
- **Secondary School Certificate** | 78.2% | Sri Vivekananda EM High School | 2008

Key Research Projects

- **Current Research Focus (Alzheimer's Disease Detection):** Researching Explainable Artificial Intelligence for Early Alzheimer's Disease Detection using multimodal MRI, deep learning, Grad-CAM, SHAP, hybrid CNN architectures, clinical decision support, and reproducible MLOps workflows.
- **Chest Disease Detection (CNN, ResNet, DenseNet):** Achieved 92% accuracy on CT/MRI datasets; integrated clinical workflows.
- **Automatic License Plate Recognition (YOLOv8, EasyOCR):** 94.21% accuracy at 45 FPS; optimized for real-time embedded systems.
- **Breast Cancer Screening (Shearlet + SVM):** Improved diagnostic sensitivity over traditional wavelet methods for microcalcification screening.

Research Interests

- Explainable AI in Healthcare - transparent ML models for medical diagnosis.
- Computer Vision and Image processing - advanced deep learning architectures for real-time object detection, image segmentation, and anomaly detection across both general and medical applications.
- MLOps for Healthcare - scalable and compliant ML pipelines for clinical settings.
- Embedded AI in Medicine - deploying models on low-resource medical devices.

Publications

- Design and Development of an Integrated Healthcare Platform Using DCNNs (Springer, 2024)
- Automatic human defense system using nanoids (IJIEMR, 2018)
- Advanced home automation and security system using IOT technology (IJEMR, 2018)

Professional Experience

Senior ML Engineer - Quantum Techworks

October 2024 - March 2026

- Managed design and deployment of intelligent automation solutions for enterprise clients, achieving 10% accuracy improvement and 25% latency reduction.
- Spearheaded research in model interpretability and explainable AI to inform executive decisions, improve clinical transparency and product strategy.
- Led cross-functional teams in developing healthcare AI platforms with focus on regulatory compliance and clinical validation.
- Mentored junior engineers and oversaw continuous integration and delivery of ML models using MLOps and Azure.

Senior ML Engineer - Intellitech Solutions

January 2023 - September 2024

- Led end-to-end research, design, and deployment of advanced ML models for healthcare platforms, including DCNN architectures for medical image analysis and an enterprise LLM-based conversational chatbot.
- Directed cross-functional teams in building automated MLOps pipelines on AKS, ensuring seamless integration with enterprise systems.
- Drove model optimization and explainability efforts, contributing to product strategy and stakeholder engagement.
- Initiated patent-worthy innovations and provided technical leadership for strategic R&D projects in AI-driven healthcare solutions.

ML Engineer - Intel TechWorks (Remote)

July 2019 - December 2022

- Developed, trained, and deployed ML models for real-time analytics, specializing in computer vision and NLP applications.
- Built and optimized data pipelines for robust feature engineering and high-quality preprocessing using Python, TensorFlow, and PyTorch.
- Collaborated with product managers and senior engineers to translate research requirements into scalable AI solutions.
- Conducted rigorous performance testing and validation, documenting methodologies and workflows for reproducibility.
- Delivered high-quality AI/ML solutions across healthcare, ALPR, image recognition, and embedded systems projects.
- Annotated and pre-processed diverse datasets; engineered features to boost model performance and generalization.
- Employed deep learning frameworks (TensorFlow, PyTorch, Keras) and OCR tools (Tesseract, EasyOCR) for custom model development.

Embedded Project Trainer - Freelancer

February 2017 - April 2018

- Conducted hands-on training sessions on microcontroller programming, RTOS, and debugging tools for academic institutions.

- Guided students in designing real-time projects with 8051, PIC microcontrollers, and sensor integrations (Zigbee, GPS, GSM, MEMS).
- Developed comprehensive curriculum for embedded systems education and practical implementation.

Associate - SCB & SBI Cards

Buzz Works Business Pvt Ltd / Innov Source Pvt Ltd

June 2016 - January 2017 / November 2014 - May 2016

- Achieved Pan India Best Employee of the Year award for exceptional performance in card operations.
- Supported process optimization and customer service excellence in financial services domain.

Embedded Project Trainer - Sarayu Software Solutions

May 2014 - October 2014

- Trained students in embedded C/C++, microcontroller interfacing, and real-time system design.
- Developed practical training modules for electronics and embedded systems engineering.

Skills

- **ML/DL Frameworks:** TensorFlow, PyTorch, Keras, LLMs, Explainable AI, CNN
- **Programming:** Python, MATLAB, Java, Embedded C/C++
- **Computer Vision & Imaging:** OpenCV, YOLOv8, EasyOCR, Tesseract, Image Segmentation, Anomaly Detection
- **NLP:** NLTK, TextBlob, tokenization, NER
- **MLOps & Tools:** MLflow, Docker, Git, Jira, Jenkins, Grafana
- **Cloud:** Azure, AWS, GCP, REST APIs, Microservices
- **Embedded Systems:** FPGA, Arduino, STM32, Beaglebone Black, 8052, PIC
- **Databases:** MySQL, SQL, MongoDB, Cassandra

Certifications

- IBM AI Engineering Professional Certificate - IBM (2020)
- AI for Medicine Specialization - Coursera (2020)

Awards & Achievements

- Pan India Best Employee of the Year - SBI Cards (2015)
- Published researcher with international peer recognition.
- Industry experiences spanning healthcare, medical AI, and embedded systems.

References

Available upon request.